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Centre number 

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Candidate signature \_\_\_\_\_

I declare this is my own work.

# INTERNATIONAL GCSE CHEMISTRY

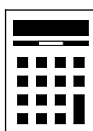
## Paper 2

Monday 11 November 2024 07:00 GMT Time allowed: 1 hour 30 minutes

### Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



### Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

| For Examiner's Use |      |
|--------------------|------|
| Question           | Mark |
| 1                  |      |
| 2                  |      |
| 3                  |      |
| 4                  |      |
| 5                  |      |
| 6                  |      |
| 7                  |      |
| 8                  |      |
| <b>TOTAL</b>       |      |





**0 1 . 3** Group 7 elements are called halogens.

Which element is a halogen?

Use **Figure 1** and the periodic table.

[1 mark]

Tick (✓) **one** box.

|          |                          |          |                          |          |                          |          |                          |          |                          |
|----------|--------------------------|----------|--------------------------|----------|--------------------------|----------|--------------------------|----------|--------------------------|
| <b>A</b> | <input type="checkbox"/> | <b>B</b> | <input type="checkbox"/> | <b>C</b> | <input type="checkbox"/> | <b>D</b> | <input type="checkbox"/> | <b>E</b> | <input type="checkbox"/> |
|----------|--------------------------|----------|--------------------------|----------|--------------------------|----------|--------------------------|----------|--------------------------|

**0 1 . 4** Which element **only** forms ions with a 1+ charge?

Use **Figure 1** and the periodic table.

[1 mark]

Tick (✓) **one** box.

|          |                          |          |                          |          |                          |          |                          |          |                          |
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|----------|--------------------------|----------|--------------------------|----------|--------------------------|----------|--------------------------|----------|--------------------------|

**0 1 . 5** Transition metals have different properties to Group 1 metals.

Which property below is shown by transition metals but **not** by Group 1 metals?

[1 mark]

Tick (✓) **one** box.

|                             |                          |
|-----------------------------|--------------------------|
| Form coloured compounds     | <input type="checkbox"/> |
| Have low densities          | <input type="checkbox"/> |
| React vigorously with water | <input type="checkbox"/> |

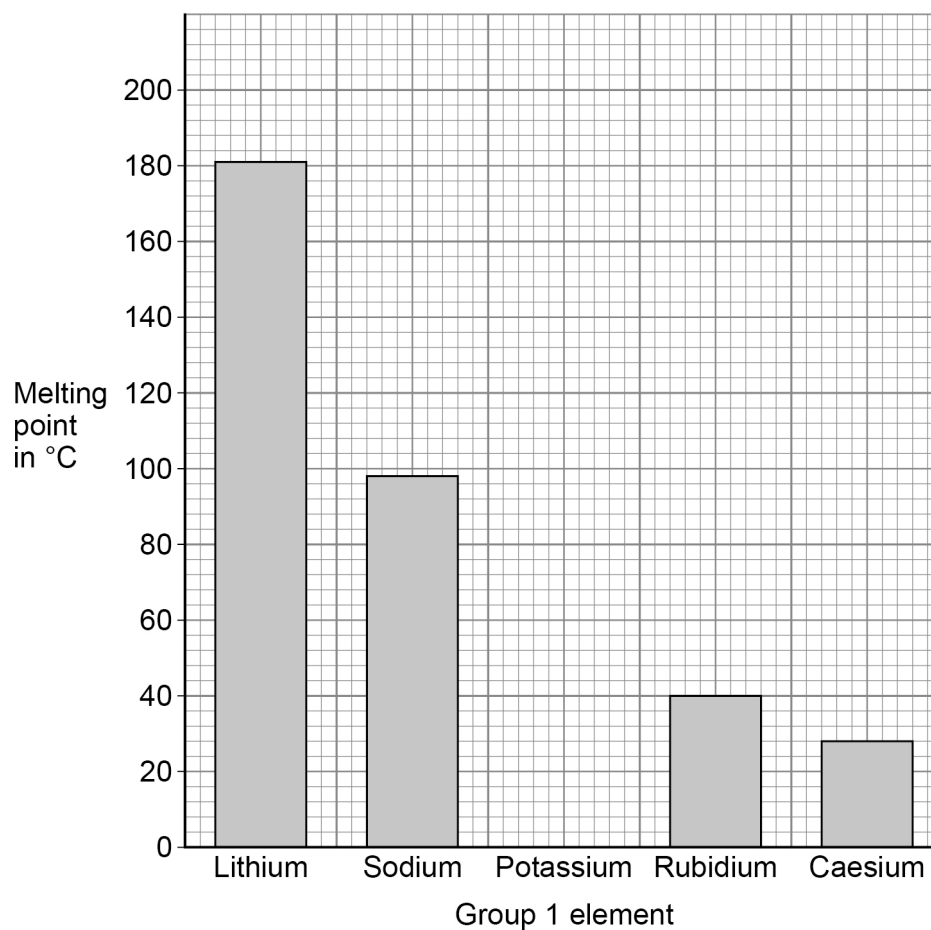
**Question 1 continues on the next page**

**Turn over ►**



**Figure 2** shows the melting points of some Group 1 elements.

**Figure 2**



**0 1 . 6** Predict the melting point of potassium.

Use **Figure 2**.

**[1 mark]**

\_\_\_\_\_ °C

**0 1 . 7** Describe the trend in melting points going down Group 1.

Use **Figure 2**.

**[1 mark]**

\_\_\_\_\_  
\_\_\_\_\_



**0 1 . 8** **Table 1** gives information about four different metals, **W**, **X**, **Y** and **Z**.

The letters **W**, **X**, **Y** and **Z** are **not** the chemical symbols for the metals.

**Table 1**

| Metal    | Charge of ion(s) formed | Catalyst |
|----------|-------------------------|----------|
| <b>W</b> | 1+                      | No       |
| <b>X</b> | 3+                      | Yes      |
| <b>Y</b> | 2+                      | No       |
| <b>Z</b> | 2+ and 3+               | Yes      |

Which metal in **Table 1** is a transition metal?

Give **two** reasons for your choice.

**[3 marks]**

Metal \_\_\_\_\_

Reason 1 \_\_\_\_\_

\_\_\_\_\_

Reason 2 \_\_\_\_\_

\_\_\_\_\_

**10**

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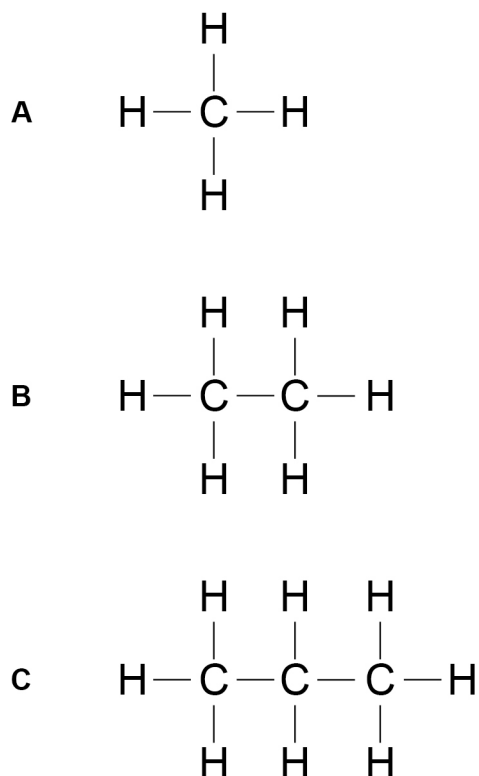
0 2

This question is about hydrocarbons.

Alkanes are hydrocarbons.

**Figure 3** shows the structure of three alkanes, **A**, **B** and **C**.

**Figure 3**



0 2 . 1

Which alkane has the **lowest** boiling point?

Use **Figure 3**.

Tick (✓) **one** box.

**[1 mark]**

**A**

☐

**B**

☐

**C**

☐


**0 2 . 2** What is the name of alkane **B**?

Use **Figure 3**.

[1 mark]

Tick (✓) **one** box.

Ethane

☐

Methane

☐

Propane

☐

**0 2 . 3** What type of bonding is present in alkanes?

[1 mark]

Tick (✓) **one** box.

Covalent

☐

Ionic

☐

Metallic

☐

**0 2 . 4** What is the general formula of an alkane?

[1 mark]

Tick (✓) **one** box.

$C_nH_n$

☐

$C_nH_{2n-2}$

☐

$C_nH_{2n}$

☐

$C_nH_{2n+2}$

☐

Question 2 continues on the next page

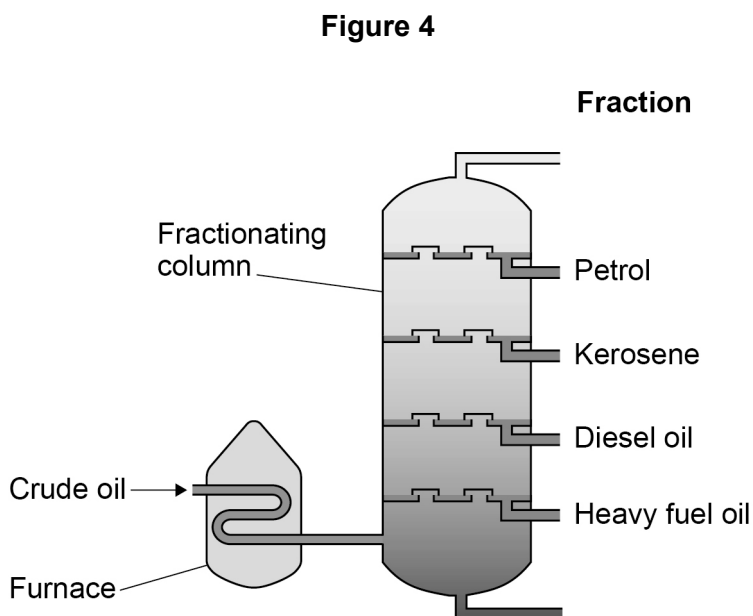
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Crude oil is a mixture of hydrocarbons.

Crude oil is separated into hydrocarbon fractions by fractional distillation.

**Figure 4** shows a fractionating column used to separate crude oil.



**0 2 5**

Complete the sentences.

Choose answers from the box.

**condense**

**evaporate**

**freeze**

**melt**

**[2 marks]**

Crude oil is heated in the furnace so the hydrocarbons \_\_\_\_\_.

The hydrocarbons go into the fractionating column.

At different temperatures in the fractionating column the hydrocarbons \_\_\_\_\_.





**0 2 . 6** Which fraction contains the smallest hydrocarbon molecules?

Use **Figure 4**.

[1 mark]

Tick (✓) **one** box.

Diesel oil

☐

Heavy fuel oil

☐

Kerosene

☐

Petrol

☐

**0 2 . 7** Which fraction has the highest viscosity?

Use **Figure 4**.

[1 mark]

Tick (✓) **one** box.

Diesel oil

☐

Heavy fuel oil

☐

Kerosene

☐

Petrol

☐

8

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**0 3**

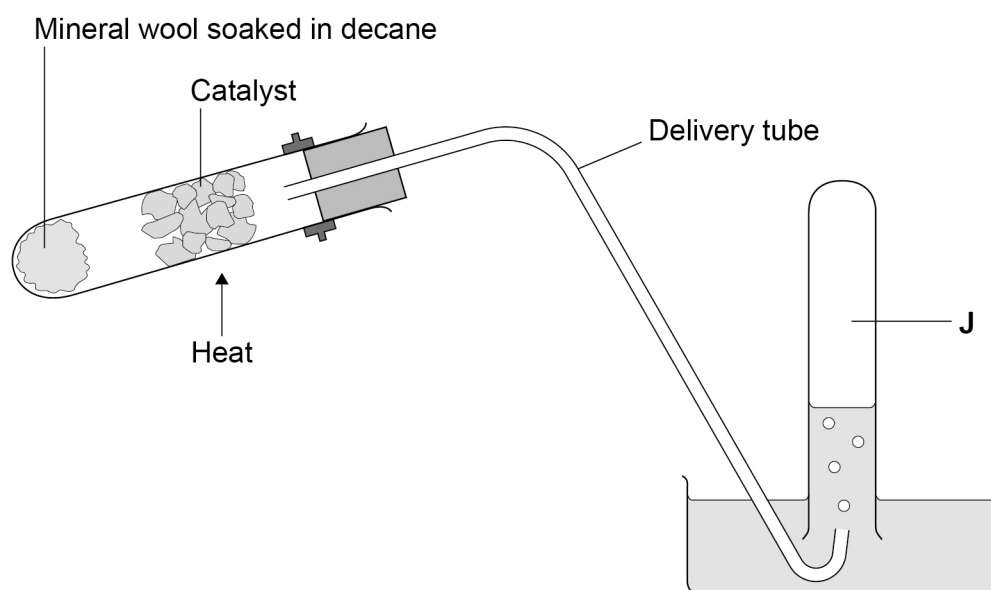
This question is about organic compounds.

Larger hydrocarbon molecules can be cracked to produce smaller hydrocarbon molecules.

A teacher demonstrated the cracking of decane ( $C_{10}H_{22}$ ).

**Figure 5** shows the apparatus.

**Figure 5**

**0 3****. 1**

Give **one** hazard of the demonstration shown in **Figure 5**.

**[1 mark]**

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**Question 3 continues on the next page**

**Turn over ►**



**0 3 . 2** What type of reaction is cracking?

**[1 mark]**

Tick (✓) **one** box.

Decomposition

☐

Displacement

☐

Neutralisation

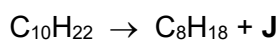
☐

Oxidation

☐

Decane is cracked to produce octane ( $C_8H_{18}$ ) and hydrocarbon **J**.

**0 3 . 3** The equation for the cracking of decane is:



What is the molecular formula of **J**?

**[1 mark]**

\_\_\_\_\_

**0 3 . 4** Hydrocarbon **J** contains a double bond.

Describe a test to show the presence of a double bond in hydrocarbon **J**.

Give the result of the test.

**[2 marks]**

Test \_\_\_\_\_

\_\_\_\_\_

Result \_\_\_\_\_

\_\_\_\_\_



**0 3 . 5** Which is a use for octane?

**[1 mark]**

Tick (✓) **one** box.

Alloy

☐

Catalyst

☐

Fuel

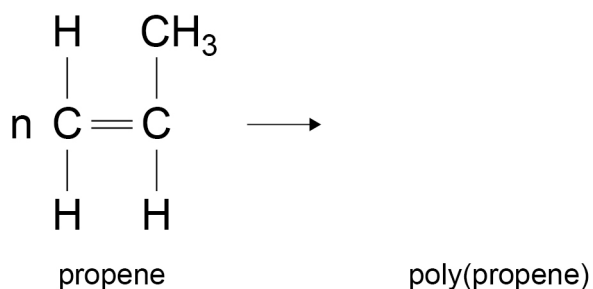
☐

The polymers poly(propene) and poly(chloroethene) are made from the products of cracking.

**0 3 . 6** Many propene molecules join together to form poly(propene).

Complete the equation to show the formation of poly(propene) from propene.

**[3 marks]**



**0 3 . 7** Poly(propene) is a thermosoftening polymer that melts when heated.

Thermosetting polymers do **not** melt when heated.

Explain why thermosetting polymers do **not** melt when heated.

**[2 marks]**

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**Question 3 continues on the next page**

**Turn over ►**



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|---|---|---|---|
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- Table 2** shows information about poly(chloroethene) and wood.

## Table 2

|                                                             | Poly(chloroethene) | Wood            |
|-------------------------------------------------------------|--------------------|-----------------|
| Raw material                                                | Crude oil          | Trees           |
| Relative mass of carbon dioxide given off during production | 487                | 457             |
| Weather resistant                                           | High               | High if treated |
| Durability                                                  | High               | Medium          |
| Biodegradability                                            | Low                | High            |

Evaluate the use of poly(chloroethene) **and** wood to make window frames.

**[6 marks]**

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17

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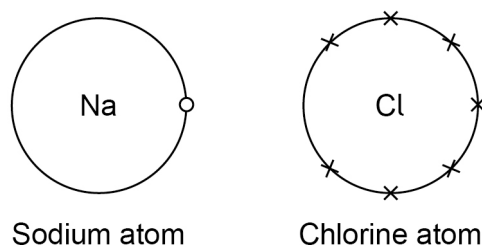
Sodium reacts with chlorine to form sodium chloride.

|   |   |   |   |
|---|---|---|---|
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|---|---|---|---|

**Figure 6** shows the electronic structure of a sodium atom and of a chlorine atom.

Only the outer shell electrons are shown.

**Figure 6**



Describe what happens when a sodium atom reacts with a chlorine atom to produce sodium chloride.

You should refer to electrons in your answer.

**[4 marks]**

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Extra space \_\_\_\_\_

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0 4 . 2

Calculate the percentage by mass of sodium in sodium chloride (NaCl).

Give your answer to 2 significant figures.

Relative atomic mass ( $A_r$ ) Na = 23Relative formula mass ( $M_r$ ) NaCl = 58.5**[3 marks]**


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Percentage by mass of sodium (2 significant figures) = \_\_\_\_\_ %

0 4 . 3

Explain why sodium chloride has a high melting point.

**[4 marks]**


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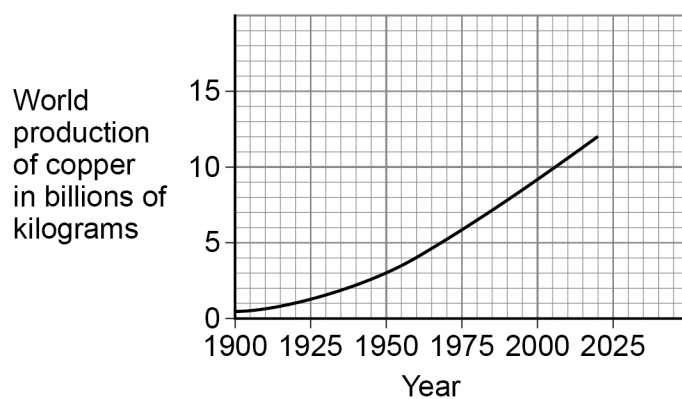
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**0 5**

This question is about methods of obtaining copper metal.

**Figure 7** shows how the world production of copper has changed.

**Figure 7**

**0 5 . 1**

Describe the trend shown by the graph in **Figure 7**.

[1 mark]

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**0 5 . 2**

Suggest **one** reason for the trend in **Figure 7**.

[1 mark]

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Copper can be extracted from copper ores using bioleaching.

Bioleaching produces leachate solutions that contain copper compounds.

Copper is then extracted from the solution of copper compounds by a displacement reaction using scrap iron.

0 5 . 3

What is used in bioleaching to produce the leachate solutions from copper ores?

[1 mark]

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0 5 . 4

Give **one** reason why iron is used to extract copper from solutions of copper compounds.

[1 mark]

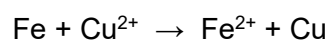
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0 5 . 5

The ionic equation for the displacement reaction is:



Explain why this is classed as a redox reaction.

[4 marks]

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**0 6**

This question is about the reaction of sulfuric acid and sodium hydroxide solution.

A student titrated  $0.20 \text{ mol/dm}^3$  sulfuric acid with sodium hydroxide solution of unknown concentration.

This is the method used:

- 1 Use a measuring cylinder to measure  $25.0 \text{ cm}^3$  of sodium hydroxide solution.
- 2 Add the sodium hydroxide solution to a conical flask.
- 3 Add 4 drops of universal indicator solution to the conical flask.
- 4 Fill a burette with  $0.20 \text{ mol/dm}^3$  sulfuric acid.
- 5 Add the sulfuric acid dropwise until the indicator changes colour.
- 6 Record the volume of sulfuric acid added.
- 7 Repeat steps **1–6** until results within  $0.10 \text{ cm}^3$  are achieved.

**0 6****1**

Suggest **one** improvement to the equipment used in step **1**.

Give **one** reason for your answer.

**[2 marks]**

Improvement \_\_\_\_\_

\_\_\_\_\_

Reason \_\_\_\_\_

\_\_\_\_\_

**0 6****2**

In step **3** the student used universal indicator.

Explain why universal indicator is **not** suitable to use in a titration.

**[2 marks]**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_



**0 6 . 3** A different student used a suitable indicator.

**Table 3** shows the results.

**Table 3**

|                                                        | <b>Titration 1</b> | <b>Titration 2</b> | <b>Titration 3</b> | <b>Titration 4</b> | <b>Titration 5</b> |
|--------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| <b>Volume of sulfuric acid added in cm<sup>3</sup></b> | 18.40              | 18.05              | 18.20              | 17.95              | 18.05              |

Calculate the mean volume of sulfuric acid added to the sodium hydroxide solution.

Do **not** include the anomalous results in your calculation.

**[3 marks]**

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Mean volume = \_\_\_\_\_ cm<sup>3</sup>

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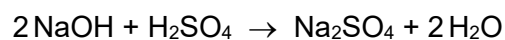
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0 6 . 4

19.05 cm<sup>3</sup> of 0.30 mol/dm<sup>3</sup> sulfuric acid completely reacted with 25.0 cm<sup>3</sup> of sodium hydroxide solution.

The equation for the reaction is



Calculate the concentration of the sodium hydroxide solution.

You should calculate:

- the number of moles of sulfuric acid in 19.05 cm<sup>3</sup> of sulfuric acid
- the number of moles of sodium hydroxide in 25.0 cm<sup>3</sup> of solution
- the concentration of sodium hydroxide solution in mol/dm<sup>3</sup>.

**[4 marks]**

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Concentration of sodium hydroxide solution = \_\_\_\_\_ mol/dm<sup>3</sup>

11



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0 7

This question is about metal carbonates.

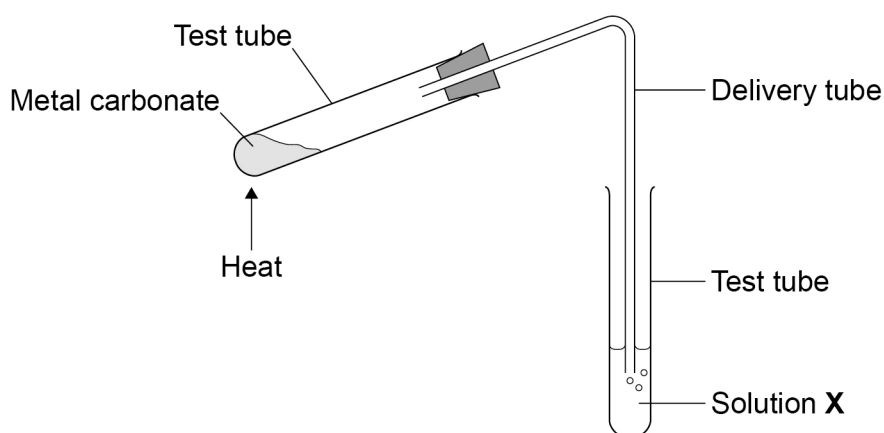
A student heated 0.5 g of a metal carbonate.

The student made observations before and after the heating.

The student repeated the process with different metal carbonates.

**Figure 8** shows the apparatus.

**Figure 8**



**Table 4** shows the results.

**Table 4**

| Metal carbonate     | Colour of metal carbonate before heating | Observations after heating |              |
|---------------------|------------------------------------------|----------------------------|--------------|
|                     |                                          | Test tube contents         | Solution X   |
| Calcium carbonate   | White solid                              | White solid                | Turns cloudy |
| Copper carbonate    | Green solid                              | Black solid                | Turns cloudy |
| Potassium carbonate | White solid                              | White solid                | No change    |





**0 7 . 1**

Solution **X** tests for the gas produced when calcium carbonate is heated.

Give the name of:

- solution **X**
- the gas produced.

Use **Figure 8** and **Table 4**.

[2 marks]

Solution **X** \_\_\_\_\_

Gas \_\_\_\_\_

**0 7 . 2**

Explain why there is a colour change when copper carbonate is heated.

Use **Table 4**.

[2 marks]

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**0 7 . 3**

Name a metal carbonate that will **not** react when heated with a Bunsen burner.

Give **one** reason why the metal carbonate does not react.

Use **Table 4**.

[2 marks]

Metal carbonate \_\_\_\_\_

Reason \_\_\_\_\_

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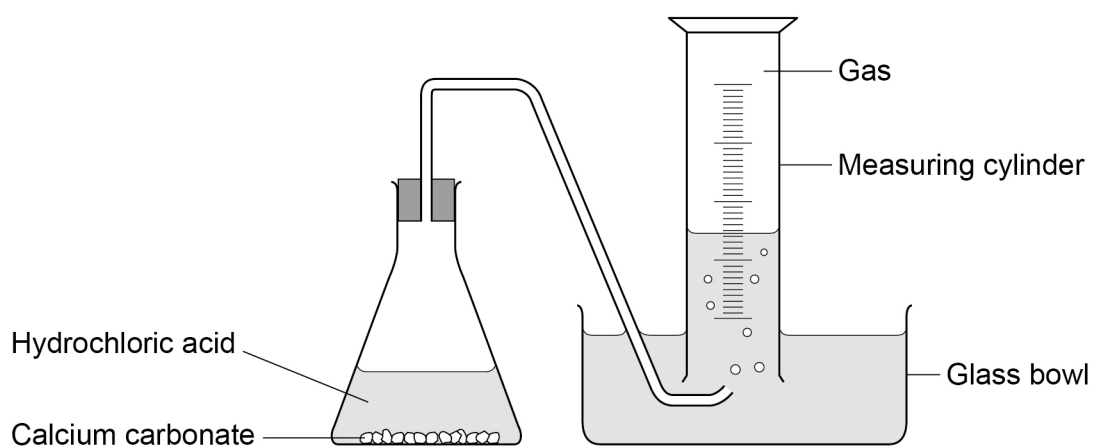


A student reacted different masses of calcium carbonate with hydrochloric acid.

The student measured the total volume of gas collected.

**Figure 9** shows the apparatus.

**Figure 9**



**Table 5** shows the results.

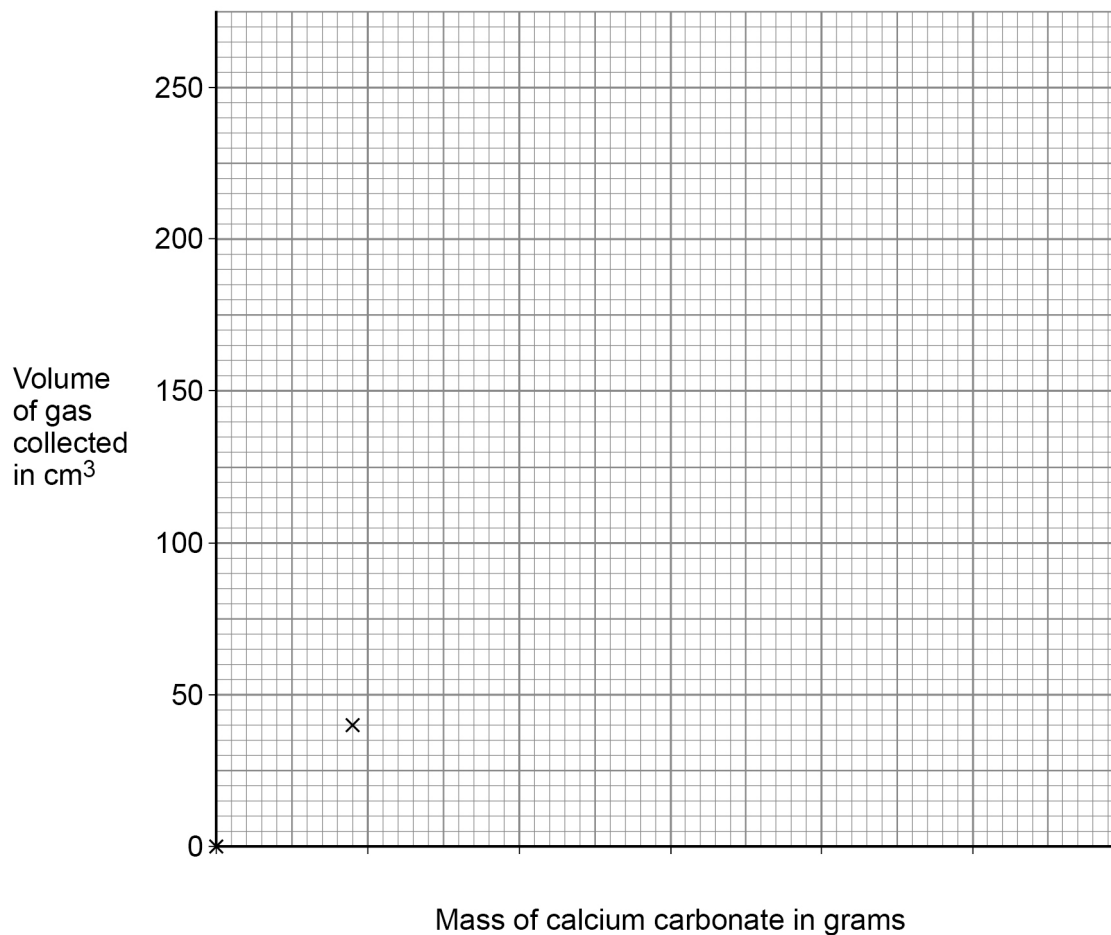
**Table 5**

| Mass of calcium carbonate<br>in grams | Volume of gas collected<br>in cm <sup>3</sup> |
|---------------------------------------|-----------------------------------------------|
| 0.00                                  | 0                                             |
| 0.18                                  | 40                                            |
| 0.35                                  | 50                                            |
| 0.53                                  | 125                                           |
| 0.71                                  | 160                                           |
| 0.88                                  | 205                                           |
| 1.06                                  | 245                                           |

0 7 . 4

**Figure 10** shows how the volume of gas collected changed as the mass of calcium carbonate reacted increased.

**Figure 10**



Complete **Figure 10**.

Some points have been plotted for you.

You should:

- use a suitable scale for the x-axis
- plot the data from **Table 5**
- draw a line of best fit.

**[4 marks]**

**Question 7 continues on the next page**



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0 7 . 5

The volume of gas collected for 0.35 g of calcium carbonate is anomalous.

Explain how the anomalous result might have occurred.

[2 marks]

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0 7 . 6

What is the **dependent** variable in this investigation?

[1 mark]

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0 7 . 7

The student repeated the investigation using hydrochloric acid at a higher temperature.

The student used 0.53 g of calcium carbonate.

Explain the effect on the volume of gas collected.

[2 marks]

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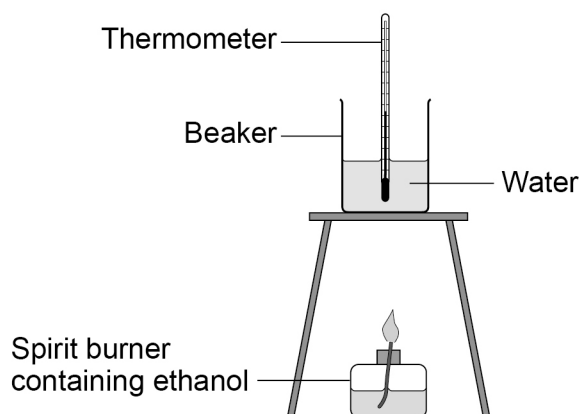


**0 8**

A student investigated the energy content of ethanol.

**Figure 11** shows the apparatus.

**Figure 11**



A student heated 100 g of water by burning ethanol in a spirit burner.

The student recorded the mass of ethanol and the temperature of the water:

- before burning the ethanol
- after burning the ethanol.

**0 8****1**

The student used a balance to measure the mass of the spirit burner and ethanol.

The balance showed a reading of 0.30 g before the spirit burner was placed on the balance.

Name the type of error shown by the balance.

**[1 mark]**

**Question 8 continues on the next page**

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**Table 6** shows the results.

### Table 6

|              | Temperature of water<br>in °C | Mass of ethanol<br>in grams |
|--------------|-------------------------------|-----------------------------|
| At the start | 18.7                          | 28.5                        |
| At the end   | 52.3                          | 26.9                        |

Determine the energy released when one gram of ethanol is burned.

Give your answer in kJ/g.

Use the equation:

$$Q = mc\Delta T$$

$c$  = specific heat capacity of water =  $4.2 \text{ J/g } ^\circ\text{C}$

**[6 marks]**

[illegible]

Energy released =                      kJ/g



08.3

Different students can do the same investigation.

Which term describes an investigation done by other students who obtain similar results?

Choose the answer from the box.

precise

repeatable

reproducible

[1 mark]

08.4

The value the student determined for the energy released by burning ethanol was much lower than the theoretical value.

Explain why the experimental value is much lower than the theoretical value.

[2 marks]

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END OF QUESTIONS



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